

Chapter 6

9. Suppose $a, b \in \mathbb{R}$. If a is rational and ab is irrational, then b is irrational.

Suppose a is rational, ab is irrational, and b is rational

$$a = \frac{k}{l}, b = \frac{m}{n}, \text{ then } ab = \frac{km}{ln} \text{ which is rational}$$

Therefore, there is no contradiction.

10. There exist no integers a and b for which $21a + 30b = 1$.

Suppose $\exists a, b \in \mathbb{Z}$, s.t. $21a + 30b = 1$

for $21a + 30b$ to equal 1, $d = \gcd(a, b) \wedge d|1$

only $1|1 \Rightarrow 21(-1) + 30(1) = 9$

Therefore, there is no contradiction.

11. There exist no integers a and b for which $18a + 6b = 1$.

Suppose $\exists a, b \in \mathbb{Z}$, s.t. $18a + 6b = 1$

$18a$ is even $\wedge 6b$ is even

Since $2|18a - 6b$ there is no contradiction.

12. For every positive $x \in \mathbb{Q}$, there is a positive $y \in \mathbb{Q}$ for which $y < x$.

Suppose $\exists a \in \mathbb{Q}^+, \forall b \in \mathbb{Q}^+, \text{ s.t. } b > a$

a would have to be the minimum of all rational numbers which does not exist, meaning this statement isn't true.

There is no contradiction

Chapter 7

7. Suppose $x, y \in \mathbb{R}$. Then $(x + y)^2 = x^2 + y^2$ if and only if $x = 0$ or $y = 0$.

$$\text{Case 1 : } (x + y)^2 = x^2 + y^2 \Rightarrow x = 0 \vee y = 0$$

$$\text{Suppose } x \neq 0 \wedge y \neq 0 \Rightarrow (x + y)^2 \neq x^2 + y^2$$

$$\text{Proof : } (x + y)^2 = (x + y)(x + y)$$

$$\rightarrow x(x + y) + y(x + y)$$

$$\rightarrow x^2 + 2xy + y^2 \neq x^2 + y^2$$

This proves the contrapositive true.

$$\text{Case 2 : } x = 0 \vee y = 0 \Rightarrow (x + y)^2 = x^2 + y^2$$

$$\text{Proof : } (x + 0)^2 = x^2 + 0^2 \rightarrow x^2 = x^2 \text{ this is true}$$

$$(0 + y)^2 = 0^2 + y^2 \rightarrow y^2 = y^2 \text{ this is also true}$$

Thus this is true.

11. Suppose $a, b \in \mathbb{Z}$. Prove that $(a - 3)b^2$ is even if and only if a is odd or b is even.

$$2|(a - 3)b^2 \Rightarrow 2|a \vee 2|b \quad \text{or} \quad 2|a \vee 2|b \Rightarrow 2|(a - 3)b^2$$

12. There exists a positive real number x for which $x^2 < \sqrt{x}$

$$\text{Proof : } 10^{-4} = (10^{-2})^2$$

$$10^{-2} = 0.01 \text{ \& } 10^{-4} = 0.0001$$

$$\text{This shows } k \in \mathbb{R}, k^{-n} < \sqrt{k^{-n}}$$

18. There is a set X for which $\mathbb{N} \in X$ and $\mathbb{N} \subseteq X$.

$$\text{Proof : } X = \mathbb{R} \Rightarrow \mathbb{N} \in X \wedge \mathbb{N} \subseteq X$$

20. There exists an $n \in \mathbb{N}$ for which $11|(2^n - 1)$.

$$\text{Case : } n = 10, 11|(2^n - 1)$$

$$\text{Proof : } 2^{10} - 1 = 1024 - 1 = 1023$$

$$11 * 93 = 90 * 11 + 3 * 11 = 990 + 33 = 1023$$

$$11 * 93 = 2^{10} - 1$$